NAME: Hairil Rashmizal Abdul Razak
DATE OF BIRTH: 2nd April 1981
PLACE OF BIRTH: Perlis, MALAYSIA.
ADDRESS:
Medical Imaging Programme,
Department of Health and Care Professions,
Faculty of Health and Life Sciences,
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G.20 South Cloisters, St Luke's Campus,
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QUALIFICATIONS:

1. RMIT University, Australia
   PhD in Medical Radiations Science (Nuclear medicine/PET-CT)
   Date of completion: December 2011
2. National University of Malaysia (UKM)
   BSc. (Hons) in Diagnostic Imaging and Radiotherapy (First Class; CGPA: 3.70/4.00) Date of completion: July 2004

PRESENT POST:

1. Senior Lecturer, Medical Imaging Programme, Department of Health and Care Professions, Faculty of Health and Life Sciences, University of Exeter (6th January 2022 – Present).
2. Senior Academic Lead, Medical Imaging Programme, Department of Health and Care Professions, Faculty of Health and Life Sciences, University of Exeter (2nd September 2024 – Present).

PAST WORKING EXPERIENCES:

1. Acting Head, Department of Nuclear Imaging, Teaching Hospital of University Putra Malaysia (HPUPM) (2nd September 2019 – 29th May 2020)
2. Acting Head, Department of Radiology, Faculty of Medicine and Health Sciences, University Putra Malaysia (UPM) (26th July 2019 – 11th September 2019)
3. Associate Professor, Faculty of Medicine and Health Sciences, University Putra Malaysia (UPM) (2nd May 2018 – 5th January 2022)
4. Deputy Dean for Research and Industrial Linkages, Faculty of Health Sciences, University Technology MARA (UiTM) (1st October 2016 – 31st March 2018)
5. Head of Centre for Graduate Studies, Faculty of Health Sciences, University Technology MARA (UiTM) (1st December 2015 – 30th September 2016)
6. Head Department of Medical Imaging, Faculty of Health Sciences, University Technology MARA (UiTM) (1st January – 30th November 2015)
7. Senior Lecturer, Faculty of Health Sciences, University Technology MARA (UiTM) (31st December 2011 – 1st May 2018).
8. Laboratory instructor, Discipline of Medical Radiations, School of Medical Sciences, RMIT University (February 2011 – August 2011).

9. Lecturer, Department of Medical Imaging, Faculty of Health Sciences, University Technology MARA (UiTM) (November 2005 – December 2011)
10. Radiographer, Mount Miriam Cancer Hospital, Penang (September 2004 – November 2005)

INTERNATIONAL / PROFESSIONAL MEMBERSHIPS:

1. The Australian Nanotechnology Network; ANN (Members)
2. IAEA Expert (2017-Present)
3. Academy of Medicine of Malaysia; Associate Member (2021-Present)
4. British Nuclear Medicine Society; Full Member (2024-Present)
5. Institute of Physics in Engineering and Medicine; Full Member (2026-Present)

PUBLICATIONS

Thesis:

1. Hairil Rashmizal Abdul Razak (PhD thesis); The Diagnosis of Extrapulmonary Tuberculosis (EPTB): Scanning Protocols and Innovative Methods for Image Quality Improvement of FDG PET/CT Imaging. Discipline of Medical Radiations, School of Medical Sciences, RMIT University, Australia (August 2011).

Book/Monograph:

1. Hairil Rashmizal Abdul Razak, Moshi Geso and Abdul Jalil Nordin. FDG PET/CT scans of Extrapulmonary Tuberculosis: Scanning Techniques & Protocols for Image Quality Improvement. ISBN: 978-3-659-19439-9. Publisher: Lambert Academic Publishing, 2012. 272 pages.

Journal articles:

1. Hishar H, Nur Nadhira, J, Wan Hamirul Bahrin WK, Saifullizan M, Anee Suryani S, Siti Aminah M, Mohd Faiz M, Nazreen W, Hairil Rashmizal AR, Muhamad Faiz O. Preliminary *In-vivo* Evaluation of [^{68}Ga]Ga-FAPI-46 in a Fibroblast Activation Protein–Negative Breast Cancer Model. Nuclear Medicine Communications. 2026:47(6):672-678.
2. Liyana S, Christopher IE, Noraniza A, Vimala B, Hairil AR. Machine Learning Technique for Managing Vulnerable People During the Covid-19 and Related Pandemic Through Telemedicine: Systematic Review, Open Challenges, and Research Directions. Neural Processing Letters. 2026:58:26.
3. Amalina Mohamed R, Mohd Hazeman Z, Fathinul Fikri AS, Wan Mazlina MS, Hairil Rashmizal AR. Assessing Cosmic Radiation Exposure Among Royal Malaysia Air Force (RMAF) Aircrews. Malaysian Journal of Medicine and Health Sciences. 2025:21(Supp 5):322-327.
4. Nurairin Syafiqah MNM, Nurul Dizyana NA, Rafidah S, Mohammad Mudzakir ZA, Hairil Rashmizal AR, Nurul Saadiah S. Protective Shielding Effectiveness for Occupational Radiation Exposure in Coronary Angiography: Evaluating the Eye Lens Safety. Malaysian Journal of Medicine and Health Sciences. 2025:21(Supp 5):316-321.
5. Naffisah MH, Siti Noorsuriani M, Nur Athirah S, Tengku Zetty Maztura TJ, Nik Mohamad Baslan Faiz MS, Hairil Rashmizal AR. Determinants of Participatory Intention in Autism Volunteer Work for Improving Quality of Life of Persons with Autism (PWA) and their Caregivers. Information Management and Business Review. 2024:16(3(I)S):727-739.

6. Hishar H, Muhamad Faiz O, Zarif NA, Hairil Rashmizal AR, Fathinul Fikri AS. Unveiling the Potential of Prostate-Specific Membrane Antigen for Precision Diagnosis and Therapy of Prostate Cancer: A Radiopharmaceutical Perspective. The Malaysian Journal of Medical Sciences. 2024:31(4):213-217.
7. Wan Mazlina MS, NA Mohd Noor, NS Mohamad Salin, Z Mohd, Z Eshak, HR Abdul Razak. The cytotoxicity effect of bismuth oxide particles synthesized hydrothermally using different reaction temperatures in vitro. The Medical Journal of Malaysia. 2024:79(Suppl 1):8-13.
8. H Murat, MKA Karim, MMA Kechik, MF Amer, HRA Razak, I Kamal, MT Chew. Internal Dose Assessment of Lymphoma 18F-FDG Positron Emission Tomography (PET) Scan. Journal of Physics: Conference Series 2623 (2023), 012003.
9. Abdul Wahab HS, Mohd Din A, Froemming GRA, Abdul Razak HR, Ab. Rahim S, Abdul Latif AM. Three-dimensional Culture Scaffolds for Osteosarcoma Cancer Stem Cells. Journal of Health and Translational Medicine (JUMMEC). 2023 (Special Issue 2):451-461.
10. Hishar H, Muhamad Faiz O, Hairil Rashmizal AR, Zainul Amiruddin Z, Fathinul Fikri AS, Mohd Azuraidi O, Loh HY, Zarif A, Jaleezah I, Mohd Hamdi Noor AH, Zaitulhusna MS. Preparation, Optimisation, and In Vitro Evaluation of [18F]AlF-NOTA-Pamidronic Acid for Bone Imaging PET. Molecules. 2022:27:7969.
11. Nurul Saadiah S,Ann Eryyna Lema TS, Noor Shafini M, Hairil Rashmizal AR, Mohamad Shahrir M. Comparing Radiation Dose Between Contrast-Enhanced and Non-Contrast-Enhanced CTAC Acquisition in 18F-FDG-PET/CT Examination. Malaysian Journal of Medicine and Health Sciences. 2022:18(Supp 15):132-139.
12. Mohamad Aminudin S, Hairil Rashmizal AR, Marianie M. Does Single Computed Tomography Attenuation Correction able to Surrogate Serial Computed Tomography Attenuation Correction in Single-Photon Emission Computed Therapy Imaging for Peptide Receptor Radionuclide Therapy Dosimetry Result? J Med Phys 2022:47:189-93.
13. Ayuni Amalina AB, Noor Shafini M, Mohd Hafizi M, Hairil Rashmizal AR, Ann Erynna LTS, Solehuddin S. Systematic review on multilevel analysis of radiation effects on bone microarchitecture. BioMed Research International. Volume 2022. Article ID: 9890633 (12 pages).
14. Izdihar K, Hairil Rashmizal AR, Muhammad Khalis AK, Syamsiah M, Josephine Ying CL, Yiin JL, Nur Atiqah Z, Norkhairunnisa M, Nur Athirah Syima MR. Mechanical and imaging properties of a clinical-grade kidney phantom based on polydimethylsiloxane and elastomer. Polymers. 2022:14:535.
15. Nur Shafinaz MS, Wan Mazlina MS, Hairil Rashmizal AR, Fatimah S. Effect of storage temperatures on physico-chemicals, phytochemicals and antioxidant properties of watermelon juice (citrullus lanatus). Metabolites. 2022:12:75.
16. Hishar H, Muhamad Faiz O, Zainul Amiruddin Z, Fathinul Fikri AS, Hairil Rashmizal AR. Multivariate optimisation and validation of the analytical GC-FID for evaluating organic solvents in radiopharmaceutical. Journal of King Saud University – Sciences. 2021:33:101554.
17. Izdihar K, Muhammad Khalis AK, Hanif Haspi H, Hairil Rashmizal AR, Low YJ, Josephine Liew YC, Mohd Mustafa AK. Evaluation of radiation attenuation properties on a various composition of polydimethylsiloxane (PDMS) for fabrication of kidney phantom. Radiation Physics and Chemistry. 2021:189:109661.

18. Muhamad Idham M, Mohd Khairul Amran M, Hairil Rashmizal AR, Wan Mazlina MS. Nephrotoxicopathology Properties of Gold and Iron Oxide Nanoparticles with Perchloric Acid & SiPEG as Radiographic Contrast Media. Malaysian Journal of Medicine and Health Sciences. 2021:17(Supp 3):92-98.
19. Hishar H, Othman MF, Abdul Razak HR. 2021. Optimal [18F]-fluorination conditions for the high radiochemical yield of [18F]AlF-NOTA-NHS complexes. Radiochimica Acta, 109(7):567-574
20. H Hassan, MF Othman, HR Abdul Razak. 18F-labeled bisphosphonate as an Alternative Candidate to the Gold Standard[18F]sodium fluoride ([18F]NaF) for PET Bone Imaging. Current Medical Imaging. 2021:17(1).
21. H Hassan, MF Othman, ZA Zakaria, FF Ahmad Saad, HR Abdul Razak. Analytical GC-FID Method for the Determination of Organic Solvents in Radiopharmaceuticals. Current Radiopharmaceuticals. 2021:14(2):131-144.
22. Nur Syahfinaz Hidayah R, Faikah Z, Hairil Rashmizal AR. Quantification of Muscle Metabolites Using Proton Magnetic Resonance Spectroscopy (1H-MRS) for Incomplete Spinal Cord Injury Patients: Preliminary Study. Malaysian Journal of Medicine and Health Sciences. 2021:17(2):4.
23. Izdihar K, Muhammad Khalis AK, Mohd Mustafa AK, Xinni N, Hairil Rashmizal AR. Evaluation of healthcare science student learning styles based VARK analysis technique. International Journal of Evaluation and Research in Education (IJERE). 2021:10(1):255.
24. H Hassan, MF Othman, ZA Zakaria, FF Ahmad Saad, HR Abdul Razak. Assessing the influence of neglected GC-FID variables on the multiple responses using multivariate optimization for the determination of ethanol and acetonitrile in radiopharmaceuticals. SN Applied Sciences. 2021:3(2):251.
25. K Izdihar, HR Abdul Razak, N Supion, MK Abdul Karim, NH Osman, M Norkhairunnisa. Structural, Mechanical, and Dielectric Properties of Polydimethylsiloxane and Silicone Elastomer for the Fabrication of Clinical-Grade Kidney Phantom. Applied Sciences. 2021:11(3):1172.
26. HH Harun, MKA Karim, Z Abbas, A Sabarudin, SC Muniandy, HRA Razak, KH Ng. The influence of iterative reconstruction level on image quality and radiation dose in CT pulmonary angiography examinations. Radiation Physics and Chemistry. 2021:178:108989.
27. MA Said, HR Abdul Razak, M Musarudin. The comparison of serial SPECT-CT imaging to estimate absorbed dose to the organ at risk from peptide receptor radionuclide therapy dosimetry. Iran J Nucl Med. 2021:29(1):2-7.
28. HH Harun, MK Abdul Karim, MA Abd Rahman, HR Abdul Razak, IN Che Isa, F Harun. Establishment of CTPA Local Diagnostic Reference Levels with Noise Magnitude as a Quality Indicator in a Tertiary Care Hospital. Diagnostics. 2020:10(9):680-690.
29. HH Harun, MKA Karim, NA Muhammad, HRA Razak, A Sabarudin, SC Muniandy. Effect of Iterative Reconstruction Algorithm Associated with Low Contrast Detectability Performance from CT Pulmonary Angiography Examinations. Journal of Physics: Conference Series 1505 (1), 012029.
30. MA Said, MA Masud, Abdul Razak HR. Quantitative imaging with commercial SPECT. Journal of Physics: Conference Series 1497 (1), 012025.
31. Aso Muhammad AM, Norliza I, Rohana A, Muhammad KA, Zamri R, Zuraiza MZ and Hairil Rashmizal AR. Effect of reconstruction parameters on cone beam CT trabecular bone microstructure quantification in sheep. BMC Oral Health. 2020:20:48.

32. Nur Syahfinaz Hidayah R, Faikah Z, Farahnaz AAB, Hairil Rashmizal AR. Proton Magnetic Resonance Spectroscopy (^{1}H-MRS) of Human Skeletal Muscle at 1.5 Tesla: Potential Applications in Exercise – A pilot Study. Malaysian Journal of Fundamental and Applied Sciences. 2019:15(6):772-775.
33. Rasdin R, Hairil Rashmizal AR, Mohd Ilham A, and Wan Mazlina MS. Supplementation of 100% Flesh Watermelon [*Citrullus lanatus (Thunb.) Matsum.* and *Nakai*] Juice Improves Swimming Performance in Rats. Preventive Nutrition and Food Science. 2019:24(1):41-48.
34. Nor Aniza A, Hairil Rashmizal AR, Vinjamuri S. PET-CT in Esophageal Cancer Management: A Cost Effectiveness Analysis Study. Malaysian Journal of Fundamental and Applied Sciences. 2019:15(1):121-124.
35. Rasdin R, Hairil Rashmizal AR, Mohd Ilham A, and Wan Mazlina MS. Separation of L-arginine and L-citrulline in Red and Yellow Crimson Watermelon (Citrullus Lanatus) Juices Extract Using HPLC Gradient Mode. Malaysian Journal of Analytical Sciences. 2018:22(5):785-793.
36. Abdul Razak HR, Azmi NA, and Vinjamuri S. Utilisation of PET-CT in Oesophageal Cancer Management: A Clinician's Perspective. Pertanika J. Sci. & Technol. 2018:26(2):821-828.
37. Md Saad WM, Mohd Nor NA, Abdul Razak HR, Mat Salleh MS, Abdul Rasid A, and Hanifah R. Watermelon Radioprotection against Radiation Stress in Mice. Pertanika J. Sci. & Technol. 2018:26(2):811-820.
38. Rasdin R, Hairil Rashmizal AR, Mohd Ilham A, and Wan Mazlina MS. Development of Isocratic RP-HPLC Method for Separation and Quantification of L-Citrulline and L-Arginine in Watermelons. International Journal of Analytical Chemistry. Volume 2018. Article ID: 4798530 (8 pages).
39. Farida Aimi M, Farahnaz AA, Ihsan MY, Fathinul Fikri AS, Abdul Jalil N and Hairil Rashmizal AR. Estimation of kidneys and urinary bladder doses based on the region of interest in 18-fluorine-fluorodeoxyglucose positron emission tomography/computed tomography examination: a preliminary study. Quantitative Imaging in Medicine and Surgery. 2017:7(3):310-317.
40. Iza Nurzawani CI, Sulaiman MD, Hairil Rashmizal AR, Ummi. Farhana H. One-Time Prolonged Ultrasound Exposure during Early Pregnancy Affects Bone Strength in Young Aged Oryctolagus Cuniculus. Pertanika J. Sci. & Technol. 2017:25(1):359-370.
41. Nur Amirah MN, Rasdin R, Nornaizie CN, Hairil Rashmizal AR, Wan Mazlina MS. *In vivo* Assessment on Acute Toxicity of Iron Oxide Nanoparticles with Different Coatings. Jurnal Teknologi. 2016:78(6-7):31-36.
42. Mohd Nazmi N, Abdul Jalil N, Fathinul Fikri AS, Hairil Rashmizal AR. Delayed Dual-Time Point Imaging Protocol Improving Myocardial Uptake of 18-Fluorine Flouro Deoxyglucose (18F-FDG) During Viability Screening of Diabetes Mellitus Type 2 Patients Using Integrated Positron Emission Tomography/Computed Tomography (PET/CT) Imaging. British Journal of Medicine & Medical Research. 2016:13(12):1-7.
43. Mohd Nazmi N, Abdul Jalil N, Fathinul Fikri AS, Hairil Rashmizal AR. Left Ventricular 2-[18F]-Fluoro-2-Deoxy-D-Glucose Uptake Using Modified Oral Glucose Loading Protocol with Pre-Medicated Niacin on Positron Emission Tomography/Computed Tomography - A Preliminary Study. Current Molecular Imaging. 2014:3(3):246-251.

44. Muhamad Idham M, Mohd Khairul Amran M, Hairil Rashmizal AR, Khairunisak AR, Wan Mazlina MS. Nanotoxic Profiling of Novel Iron Oxide Nanoparticles Functionalized with Perchloric Acid and SiPEG as a Radiographic Contrast Medium. BioMed Research International. Volume 2015. Article ID: 183525 (7 pages).
45. Sun Z, Rashmizal H, Xu L. Molecular imaging of plaques in coronary arteries with PET and SPECT. J Geriatr Cardiol. 2014:11:E16-E29.
46. Mohd Khairul Amran M, Muhamad Idham M, Ainul Mardhiyah Z, Hairil Rashmizal AR, Wan Mazlina MS. Watermelon [citrullus lanatus (thunb.) matsum. and nakai] juice modulates oxidative damage induced by low dose x-ray in ICR mice. BioMed Research International. (Special Issue on Biologic Activity and Biotechnological Development of Natural Products). Volume 2014. Article ID: 512834 (6 pages).
47. Muhamad Idham M, Mohd Khairul Amran M, Ainul Mardhiyah Z, Nurhanisa G, Mazriazie MI, Hairil Rashmizal AR, Wan Mazlina MS. Induction of oxidative stress following low dose ionizing radiation in ICR mice. World J Med Sci. 2014:10(2):198-203.
48. Ainul Mardhiyah Z, Nurhanisa G, Mohd Khairul Amran M, Muhamad Idham M, Mazriazie MI, Hairil Rashmizal AR, Wan Mazlina MS. Radioprotective effect of watermelon juice against low dose ionizing radiation-induced inflammatory response in mice. World J Med Sci. 2014:10(2):191-197.
49. Hairil Rashmizal AR, Said Mohd Shaffiq SR, Wan Mazlina MS. Effects of different tube potentials and iodine concentrations on image enhancement, contrast-to-noise ratio and noise in micro-CT images: a phantom study. Quant Imaging Med Surg. 2013:3(5):256-261.
50. Farah Wahida AZ, Sulaiman MD, Hairil Rashmizal AR, Hamzah Fansuri H. Prenatal ultrasound heating impacts on fluctuations in haematological analysis of *Oryctolagus cuniculus*. Quant Imaging Med Surg. 2013:3(5):262-268.
51. Sulaiman MD, Hairil Rashmizal AR, Farah Wahida AZ, Nurul Hidayah S, Khairunnisa AM, Iza Nurzawani CI and Ummi Farhana H. Ultrasound exposure during pregnancy affects rabbit foetal parathyroid hormone (PTH) level. Quant Imaging Med Surg. 2013:3(1):49-53.
52. Hairil Rashmizal AR, Geso M, Noraini AR and Abdul Jalil N. Imaging characteristics of extrapulmonary tuberculous lesions on dual time point imaging (DTPI) of FDG PET/CT. Journal of Medical Imaging and Radiation Oncology. 2011:55(6):556-562.
53. Abdul Razak HR, Nordin AJ, Ackerly T, Van Every B, Martin R and Geso M. Quantifying the effects of iodine contrast media on standardised uptake values of FDG PET/CT images: an anthropomorphic phantom study. Australas Phys Eng Sci Med. 2011:34(3):367-374.
54. Abdul Jalil N, Suraini MS, Hairil Rashmizal AR and Noraini AR. Multimodality Diagnostic Imaging in Tuberculous Lymphadenitis: A Case Review. Pertanika J. Sci. & Technol. 2011;19(1):141-146.
55. Hairil Rashmizal AR, Geso M, Noraini AR and Abdul Jalil N. The effect of integrating contrast enhanced CT in FDG PET/CT imaging of Extrapulmonary Tuberculosis patients: preliminary data. ANZ Nuclear Medicine. 2010;41(4):19-23.
56. Hairil Rashmizal AR, Moshi Geso, Noraini AR and Abdul Jalil N. Positron Emission Tomography/Computed Tomography (PET/CT) can be Used to Differentiate between Tuberculosis and Cancer Where Other Modalities are Challenged. SGH Med Jour. 2010:4(2):89-92.
57. HR Abdul Razak, N Abdul Rahim and AJ Nordin. Dual time point imaging of FDG PET/CT in a tuberculous spondylodiscitis. Biomed Imaging Interv J. 2010;6(2):e18.

58. Hairil Rashmizal AR, Noraini AR, C Rossetti and Abdul Jalil N. Case report: Brown fat uptake of ^{18}F-FDG on dual time point PET/CT imaging. Singapore Med J. 2010;51(2):e37-e39.
59. Hairil Rashmizal AR, Noraini AR and Abdul Jalil N. Dual time point imaging ^{18}F-FDG PET/CT in a cold tuberculous abscess: a case report. SGH Med Jour. 2010;4(1):55-59.
60. Nordin AJ, Rashmizal H, Abdul Rahim N and Rossetti C. Potential false positive active extra pulmonary tuberculosis lesions on FDG PET/CT imaging in malignancy. Dicle Med J. 2010;37(1):42-47.

INVITED SPEAKER / EXPERT:

1. Invited speaker for CME online organised by Technical Committee of Diagnostic Radiographers Profession, Allied Health Sciences Division, Ministry of Health Malaysia; 17th April 2021.
2. Invited speaker for Webinar Series in Imaging Sciences, College of Health Sciences, Gulf Medical University, UAE; 10th April 2021.
3. Invited speaker for Workshop Series Radiographers for Research (R4R), Malaysian Society of Radiographers (MSR); 30 – 1st December 2019; UnisZA Gong Badak Campus, Terengganu.
4. Invited speaker for 1st Malaysian Radiographers Conference (MDRC); 24 - 25th August 2019; The Everly Hotel, Putrajaya.
5. Invited speaker for Basic Research for Healthcare Profesional Bootcamp; 29th June 2019; Computer Lab, Faculty of Health Sciences, Universiti Kebangsaan Malaysia (UKM), Kuala Lumpur Campus.
6. Invited speaker for Research Writing Workshop; 22nd April 2019; Seminar Hall, Faculty of Health Sciences, Universiti Teknologi MARA (UiTM) Selangor Branch, Puncak Alam Campus, Selangor.
7. Invited speaker for National Radiology Transplant Symposium; 10-12th April 2019; AKEPT, Bandar Enstek, Negeri Sembilan.
8. Invited speaker for Medical Imaging Research Colloquium; 9th July 2018; Seminar Hall, Faculty of Health Sciences, Universiti Teknologi MARA (UiTM) Selangor Branch, Puncak Alam Campus, Selangor.
9. Invited speaker for PET/CT Training Course I; 4 – 5th December 2017; Seminar Room PPDN, Faculty of Medicine & Health Sciences, Universiti Putra Malaysia, Serdang, Selangor.
10. IAEA Expert Mission to Kuwait; Jaber Al Ahmad Nuclear Medicine and Molecular Imaging Center (JANMMIC), Kuwait, 20 – 22nd March 2017.
11. Invited speaker for PET/CT Training Course II; 13 – 14th November 2014; Seminar Room PPDN, Faculty of Medicine & Health Sciences, Universiti Putra Malaysia, Serdang, Selangor.
12. Invited speaker for Continuous Medical Education (CME) on Standardization of PET/CT Imaging Technique; 8th April 2014; Library Auditorium, Faculty of Medicine & Health Sciences, Universiti Putra Malaysia, Serdang, Selangor
13. Invited speaker for Seminar on Radiation Protection in Medical Imaging and Environmental Health 2013; 7 – 8th September 2013; Serdang Hospital's Auditorium, Selangor.
14. Invited speaker for National Seminar on Medical Radiation 2012; Transdisciplinary Approach in Advancement of Cancer Research; 15th April 2012; School of Health Sciences, Universiti Sains Malaysia, Kubang Kerian, Kelantan.

POSTGRADUATE RESEARCH SUPERVISION

Doctor of philosophy; PhD:

1. Tengku Muhammad Yoshandi. Department of Health and Care Professions, Faculty of Health and Life Science, University of Exeter. Fabrication of Polymer Gel Dosimeter for 3D Dose Mapping in Nuclear Medicine. Main Supervisor (On-going).
2. Thomas York. Department of Health and Care Professions, Faculty of Health and Life Science, University of Exeter. Utilising Artificial Intelligence Decision-support Tools to Supplement Clinician Judgement in Lower Limb Musculoskeletal Radiographs. Co-supervisor (**Completed**).
3. Hissa Mohammed. Department of Health and Care Professions, Faculty of Health and Life Science, University of Exeter. Exploring Changes in Brain Metabolism using PET/CT and fMRI in Long Covid Sufferers. Main Supervisor (On-going).
4. Taiba Alsaleh. Department of Health and Care Professions, Faculty of Health and Life Science, University of Exeter. Exploration of the Influence of Artificial Intelligence (AI) Implementation in CT Practice in Kuwait Public Hospitals. Main Supervisor (On-going).
5. Anwaar Asiri. Department of Health and Care Professions, Faculty of Health and Life Science, University of Exeter. Scanning Protocols Optimisation and Image Quality Improvement in FDG PET/CT Brain Imaging. Main Supervisor (On-going).
6. Amani Mansour Alsaeedi. Department of Health and Care Professions, Faculty of Health and Life Sciences, University of Exeter. Quantification of Information Loss in Fusion PET/CT Imaging Technique. Main Supervisor (On-going).
7. Bandar Almatrafi. Department of Health and Care Professions, Faculty of Health and Life Sciences, University of Exeter. Fluorocholine PET/CT Imaging in Hyperparathyroidism. Co-supervisor (On-going).
8. Muhammad Hishar Hassan. Faculty of Medicine and Health Sciences, UPM. Radiolabelling of CXCR4 Analog Using 18F-Aluminum Fluoride (18F-ALF) for Positron Emission Tomography Imaging of Chemokine Receptor. Main Supervisor (**Completed**).
9. Mohamad Aminudin Said. Faculty of Medicine and Health Sciences, UPM. Novel Technique of SPECT Lu-177 Therapy Dosimetry. Main Supervisor (**Completed**).
10. Hanif Haspi Harun. Faculty of Science, UPM. Effect of Tube Potential and Reconstruction Kernel on Low Contrast Detectability of CT Pulmonary Angiography Examination. Co-supervisor (**Completed**)
11. Izdihar Kamal. Faculty of Medicine and Health Sciences, UPM. Development of Renal Mass Phantoms for Computed Tomography Texture Analysis (CTTA). Co-supervisor (**Completed**)
12. Norhidayah Hashim. Faculty of Health Sciences, UiTM. Quantitative Evaluation of Ultrasonic Echogenicity Features of Breast Tumours for Dense Breast Women. Co-supervisor (**Completed**)
13. Nor Aniza Azmi. Faculty of Health Sciences, UiTM. PET/CT Impact in Oesophageal Cancer Management: Assessment on Practicality, Clinical Efficacy and Cost-effectivness in North West Practices. Main Supervisor (**Completed**).
14. Iza Nurzawani Che Isa. Faculty of Health Sciences, UiTM. Prenatal Ultrasound Heating Induces Early Physiological Changes Related to Bone in Newborn Rabbits. Co-supervisor (**Completed**).
15. Rashidah Shahruddin. Faculty of Health Sciences, UiTM. Correlation between Overexpression of p53: Clinical and Demographic Risk Factors among Female Breast Cancer Population. Co-supervisor (**Completed**).

Master by research:

1. Ayuni Amalina Abu Bakar. Faculty of Health Sciences, UiTM. Multilevel Analysis of Radiation Effect on Bone Morphology and Mechanical Properties. Co-supervisor (**Completed**).
2. Nur Syahfinaz Hidayah Rusli. Faculty of Health Sciences, UiTM. Lower Limb Muscles Assessment in Incomplete Spinal Cord Injury (SCI) Patients by Using Magnetic Resonance Spectroscopy (MRS). Co-supervisor (**Completed**).
3. Farida Aimi Mustapa. Faculty of Health Sciences, UiTM. Modelling FDG Radiation Dose to Kidneys. Main supervisor (**Completed**).
4. Rasdin Ridwan. Faculty of Health Sciences, UiTM. Isocratic RP-HPLC Method for Quantification of L-Citrulline Content in 100% Watermelon [Citrullus lanatus (Thunb.) Matsui & Nakai] Juices and Its Role in Improving Swimming Performance. Co-supervisor (**Completed**).
5. Mohd Nazmi Nordin. Nuclear Diagnostic Imaging Centre, UPM. Quantifying FDG Myocardial Uptake in High Fatty Acid Environment through Oral Supplement during PET/CT Assessment. Co-supervisor (**Completed**).
6. Muhamad Idham Mohamed. Faculty of Health Sciences, UiTM. Toxicity Assessment of Nanoparticles in Wistar Rat. Co-supervisor **(Completed)**.
7. Mohd Khairul Amran Mohammad. Faculty of Health Science, UiTM. Watermelon [Citrullus Lanatus (Thumb.) Matsum & Nakai] Juice Modulates Genotoxicity of Low Dose Ionizing Radiation in Mice. Co-supervisor. **(Completed)**.

POSTGRADUATE THESIS EXAMINER:

Master by research:

1. External Examiner; Knowledge, Attitude and Practice (KAP) of Radiation Safety among X-ray Radiographers in the Government Hospital in Taian City; Faculty of Health Sciences, UiTM (October 2023).
2. External Examiner; Resting Brain Responses in Precuneus and Supramarginal Gyrus: Comparison between Healthy Subjects and Temporal Lobe Epileptic (TLE) Patients; Faculty of Medicine, UniSZA (March 2021).
3. External Examiner; Benchmarking the Quality Assurance Program (QAP) Standard of Safety and Performance Tests for Linear Accelerator (LINAC); School of Medical Sciences, USM (September 2020).
4. External Examiner; Dosimetric Verification Using Optically Stimulated Luminescence (OSL) for Chest X-ray Examinations in Perak Health Clinics; School of Physics, USM (August 2020).
5. External Examiner; Estimation of Cumulative Effective Dose and Potential Risks of Radiation Exposure from Multiple Scans of CT Head; Advanced Medical and Dental Institute, USM (August 2020).
6. External Examiner; Evaluation of Contrast Enhancement Using Different Protocols for Routine Thorax Computed Tomography on 16-slice Scanners in Two Tertiary Hospitals; Advanced Medical and Dental Institute, USM, (August 2020).
7. External Examiner; Utilising Virtual Bolus in Dose Optimisation (Tomotherapy) for Head and Neck Cancers; School of Physics, USM (March 2020).
8. External Examiner; Evaluation of Radiation Dose and Image Quality of CT Imaging for Different Phantom Sizes; Faculty of Applied Sciences, UiTM (January 2020).
9. External Examiner; Enhancement of Low-level Laser Irradiation Cytotoxicity in Human Breast Cancer MCF-7 Cells by Using Gold Nanoparticles (*In-vitro* Study); School of Physics, USM (December 2019).

10. External Examiner; Comparative Study of Metal Artefact Reduction by Monoenergetic Extrapolation Application Between Different Dual Energy Computed Tomography (DECT) Techniques; Advanced Medical and Dental Institute, USM (April 2019).
11. External Examiner; Diode Dosimetric Characteristics Assessment for In-vivo Dosimetry in Radiotherapy Treatment; School of Physics, USM (September 2018).
12. External Examiner; Fabrication of Anthropomorphic Thyroid-Neck Phantom for Internal Radiation Dosimetry in Nuclear Medicine; Advanced Medical and Dental Institute, USM (March 2018).
13. External Examiner; F-18-labelled Choline: Development of Convenient Quality Control Analysis Method and Improvement on the Radiochemical Yield; Faculty of Medicine and Health Sciences, UPM (July 2016).
14. Internal Examiner; Quantitative Correlation between Mammographic Breast Density according to BI-RADs and Ultrasound Echo Intensity; Faculty of Health Sciences, UiTM (July 2013).
15. Internal Examiner; Dose Comparison in Mammography Based on Digital Mammography Database Establishment; Faculty of Health Sciences, UiTM (October 2012).

Doctor of philosophy; PhD:

1. External Examiner; Advanced Radiotherapy Treatment Plan Verification by Using EPID-based In Vivo Dosimetry; School of Physics, USM (October 2025).
2. External Examiner; Quantification of Hip Prosthetic Metal Artefacts on Different CT Scanners using a Customised Pelvic Phantom; Advanced Medical and Dental Institute, USM (October 2025).
3. External Examiner; Assessment of Risk and Image Quality Improvement with Hybrid Iterative in Brain CT Imaging; Advanced Medical and Dental Institute, USM (July 2025).
4. External Examiner; Fabrication of Semiconductors/Metal/Semiconductors Multilayer Structures for Thermoluminescence Dosimetry Application; School of Physics, USM (April 2025).
5. External Examiner; Fabrication and Evaluation of PMMAG/CuO Polymer Gel Dosimeter for Radiotherapy Applications; School of Physics, USM (March 2025).
6. External Examiner; Development of Improved Algorithm in Determination of Individual Specific-Size for Accurate CT Dose Estimation; Advanced Medical and Dental Institute, USM (February 2025).
7. External Examiner; Lead-free Metal Added Composite Particleboard for Radiation Shielding; School of Physics, USM (September 2024).
8. Internal Examiner; Simulation and Optimisation of Total Knee Replacement Imaging with Positron Emission Tomography/ Computed Tomography; Faculty of Health and Life Sciences, University of Exeter (June 2024).
9. External Examiner; Theranostic Investigation on Gadolinium-159 (159Gd) as Potential Radioisotope for Hepatic Radioembolisation Using Monte Carlo Simulations; School of Physics, USM (March 2024).
10. External Examiner; Ultrasound Assisted Bismuth Coated Iron Oxide Nanoparticles for Dual-Modal CT/MR Imaging Application; School of Physics, USM (August 2023).
11. External Examiner; Synthesis and Evaluation of Poly(ethylene) Oxide/Erbium Oxide as Diagnostic Nanofibres for Magnetic Resonance Imaging; School of Physics, USM (July 2023).

12. External Examiner; Radiation Dose Assessment in Differentiated Thyroid Cancer Patients Treated with High Dose Radioiodine Therapy Using 3D Quantitative SPECT/CT Imaging; Advanced Medical and Dental Institute, USM (February 2023).
13. External Examiner; Optimization of CT Scanning Protocol of Type B Aortic Dissection Follow-up Through 3D Printed Model; Curtin Medical School, Curtin University, Australia (February 2023).
14. External Examiner; A Study on Microcalcification of Full Field Digital Mammography Breast Phantom Images Using MatLab Algorithms; School of Physics, USM (September 2022).
15. Internal Examiner; The Clinical Utility of Digital and Long field-of-view Positron Emission Tomography for Oncology; College of Medicine and Health, University of Exeter (September 2022).
16. External Examiner; Development and Characterisation of Polymer Gel Dosimeter Doped with Gold Nanoparticles; School of Physics, USM (August 2022).
17. External Examiner; Port-processing Imaging Techniques to Enhance the Quality of Brain MRI Image; School of Physics, USM (July 2022).
18. External Examiner; Development of an Automated Calculation Method for CT Dose Estimation Based on Individual Specific Size in Paediatric Population; Advanced Medical and Dental Institute, USM (February 2022).
19. Internal Examiner; Modelling of a Scintillation Camera with Silicone-Photomultiplier (SIPM) Photo Based Sensor; Faculty of Engineering, UPM (October 2021).
20. External Examiner; Radiobiological Modelling of Gold Nanoparticles Radiosensitisation Effects in Conventional and Advanced Radiotherapy Techniques; School of Health Sciences, USM (January 2021).
21. External Examiner; Efficacy and Compatibility of Hepatocellular Carcinoma Dynamic Liver Phantom for Use in Magnetic Resonance Imaging; School of Physics, USM (October 2020).
22. External Examiner; Can Shear Wave Elastography be of Value in the Assessment of Cervical Strength for the Prediction of Preterm Birth?; School of Molecular and Life Sciences, Curtin University, Australia (May 2020).
23. External Examiner; Zinc Oxide Nanoparticles Cytotoxicity and Their Photodynamic Therapy towards Urinary Bladder Cancer Cell Lines; School of Physics, USM (October 2019).
24. External Examiner; The Effects of Cell Age on Cell-Nanoparticles Interactions; School of Physics, USM (August 2019).
25. External Examiner; Diagnostic Value of Multi-Slice CT Pulmonary Angiography in the Diagnosis of Pulmonary Embolism: An Investigation of Optimal Scanning Protocols in terms of Image Quality, Contrast and Radiation Doses; School of Molecular and Life Sciences, Curtin University, Australia (February 2019).
26. External Examiner; Diagnostic Value of Nuclear Cardiology in Coronary Artery Disease; Discipline of Medical Imaging Science, Curtin University, Australia (February 2014).

ACADEMIC / EXPERT APPOINTMENT:

1. Panel Selection Committee for the Health and Care Professions (HCP) Doctoral Award, National Institute for Health and Care Research (NIHR), UK (November 2023 – present).

2. Evaluator for Malaysia-France Bilateral Research Collaboration (MATCH) 2021 Research Grant, Department of Higher Education, Ministry of Education Malaysia (April 2021)
3. Member of Research Grants Evaluation Committee for Ministry of Higher Education Malaysia (Mac 2020 – December 2021)
4. Evaluator for Hubert Curien Partnership – Hibiscus (PHC-Hibiscus) 2020 Research Grant, Department of Higher Education, Ministry of Education Malaysia (Mac 2020)
5. Evaluator for Fundamental Research Grant Scheme (FRGS) Cycle 1/2020, National Defence University of Malaysia (UPNM) (Mac 2020)
6. Advisory Panel for Workshop Series Radiographers for Research (R4R), Malaysian Society of Radiographers (MSR) (November 2019 – October 2020)
7. National Regional Cooperative Agreement (RCA) Representative (Alternate Member) for Strengthening Capacity to Manage Non-Communicable Diseases Using Imaging Modalities in Radiology and Nuclear Medicine project (RAS6093), International Atomic Energy Agency (IAEA) (January 2018 – December 2021).
8. External Academic Advisor, Bachelor of Medical Imaging, School of Medical Imaging, Faculty of Health Sciences, University Sultan Zainal Abidin; UniSZA (November 2019 – February 2022).
9. External Academic Advisor, Bachelor of Medical Imaging Program, Department of Health Sciences, Faculty of Engineering and Life Sciences, University Selangor; Unisel (May 2018 – May 2020).
10. International Atomic Energy Agency (IAEA) Expert (March 2017 – Feb 2020).
11. External Academic Advisor, Department of Medical Imaging, Faculty of Health and Sport Sciences, MAHSA University (August 2016 – November 2022).
12. Expert Panel for Radiotherapy Field, Malaysian Qualifications Agency (MQA) (September 2012 – August 2015)
13. Editorial Board for Quantitative Imaging in Medicine and Surgery journal (QIMS; Print ISSN 2223-4292; Online ISSN 2223-4306) [*Indexed in SCIE, SCOPUS, PubMed and PubMed Central, Publisher: AME Publishing Company*].
14. Editorial Board for Journal of Cardiology & Diagnostics Research [*Non-indexed*].

MANUSCRIPT REVIEWER:

1. Medical Journal of Malaysia (Scopus Indexed)
2. Quantitative Imaging in Medicine and Surgery journal (WoS Q2)
3. Medical & Biological Engineering & Computing journal (WoS Q3)
4. Radiation Physics and Chemistry journal (WoS Q1)
5. International Journal of Radiation Research (WoS Q4)
6. Malaysian Journal of Medicine & Health Sciences (Scopus Indexed)
7. Malaysian Journal of Medical Sciences (Scopus Indexed)
8. BMC Medical Imaging journal (WoS Q3)
9. Australasian Physical & Engineering Sciences in Medicine journal (WoS Q4)
10. Journal of Mechanics in Medicine and Biology (WoS Q4)
11. Radiography (Scopus Indexed)

RESEARCH GRANTS

University level (Malaysia):

1. DUCS-Fakulti (UiTM)
   Topic: Exploring Quality of Life in Working Parents/Caregivers of Children and Adolescents with Autistic Spectrum Disorder
   Role: Member
   Amount: MYR 20,000 (~USD 4,732)
   Duration: 2 years (1 October 2022 – 30 September 2024) [**Completed**]
2. GPK; Geran Penyelidikan Khas (UiTM)
   Topic: Watermelon Profiling and UV-rays Induced Oxidative Damage
   Role: Member
   Amount: MYR 20,000 (~USD 4,732)
   Duration: 2 years (21 December 2020 – 20 December 2022) [**Completed**]
3. GPK; Geran Penyelidikan Khas (UiTM)
   Topic: Impact of Ionising Radiation Therapy on Bone Tissue Properties in the Absence of Muscle and Disease Effects: A Novel Influence
   Role: Member
   Amount: MYR 20,000 (~USD 4,732)
   Duration: 2 years (21 December 2020 – 20 December 2022) [**Completed**]
4. GP-IPS; Geran Inisiatif Putra Siswazah (UPM)
   Topic: Development of Quality Phantom for Computed Tomography Texture Analysis (CTTA) of Solid Renal Masses
   Role: Member
   Amount: MYR 19,000 (~USD 4,590)
   Duration: 3 years (17 February 2020 – 16 February 2023) [**Completed**]
5. Bestari (UiTM)
   Topic: Optimisation of CMOS-based Optical CT Imaging System for Radiotherapy Dosimetry
   Role: Member
   Amount: MYR 50,000 (~USD 12,500)
   Duration: 2 years (March 2017 – February 2019) **[Completed]**
6. Bestari (UiTM)
   Topic: Safety and Efficacy Studies of *Andrographis Paniculata* (Hempedu Bumi) for Cardiovascular Control
   Role: Member
   Amount: MYR 40,000 (~USD 10,000)
   Duration: 2 years (March 2017 – February 2019) **[Completed]**
7. GIP; Geran Inisiatif Penyelidikan) (UiTM)
   Topic: Utilisation of PET-CT in Oesophageal Cancer Management: A Comprehensive Cost Effectiveness Analysis (CEA)
   Role: Project Leader
   Amount: MYR 30,000 (~USD 7,500)
   Duration: 1 year (September 2016 – August 2017) **[Completed]**
8. Lestari (UiTM)
   Topic: PET-CT in Oesophageal Cancer Management: A Cost Effectiveness Analysis (CEA)
   Role: Project Leader
   Amount: MYR 20,000 (~USD 5,000)
   Duration: 2 years (April 2016 – March 2018) **[Completed]**

9. Lestari (UiTM)
   Topic: Culinary Internship: Assessment of Food Hygiene and Safety among Bumiputera Students
   Role: Member
   Amount: MYR 20,000 (~USD 5,000)
   Duration: 2 years (April 2016 – March 2018) **[Completed]**
10. Lestari (UiTM)
    Topic: An Evaluation of a Nutritional Educational Programme for Obese Children in Primary Schools
    Role: Member
    Amount: MYR 20,000 (~USD 5,000)
    Duration: 2 years (April 2016 – March 2018) **[Completed]**
11. Lestari (UiTM)
    Topic: Biocompatibility of Noble Metallic Nanoparticles in the Body
    Role: Member
    Amount: MYR 20,000 (~USD 5,000)
    Duration: 2 years (December 2015 – November 2017) **[Completed]**
12. Research Intensive Faculty (RIF) (UiTM)
    Topic: Inflammatory Response Following Exposures to the Annual Dose Limit for Whole Body Irradiation of Radiation Workers: an *In-vivo* Study
    Role: Project Leader
    Amount: MYR 32,000 (~USD 10,000)
    Duration: 2 years (September 2012 – August 2014) **[Completed]**
13. Research Intensive Faculty (RIF) (UiTM)
    Topic: Induction of Oxidative Stress Following Exposure to Low Dose Ionizing Irradiation
    Role: Member
    Amount: MYR 32,000 (~USD 10,000)
    Duration: 2 years (September 2012 – August 2014) **[Completed]**
14. Research Intensive Faculty (RIF) (UiTM)
    Topic: The Effect of Prenatal Ultrasound Heating on Physiological Development in *Oryctolagus Cuniculus*
    Role: Member
    Amount: MYR 32,000 (~USD 10,000)
    Duration: 2 years (July 2012 – June 2014) **[Completed]**

National level (Malaysia):

1. Fundamental Research Grant Scheme for Research Acculturation of Early Career Researchers (FRGS-RACER)
   Topic: The Efficacy of Thymoquinone Nanoparticles and Hyperthermia on Modulating Chemoresistance of Cancer Stem Cells in Osteosarcoma
   Role: Mentor
   Amount: MYR 51,200 (~USD 12,360)
   Duration: 2 years (September 2019 – August 2021) [**Completed**]

2. Fundamental Research Grant Scheme for Research Acculturation of Early Career Researchers (FRGS-RACER)
   Topic: Multi-level Analysis of Radiation Effects on Bone Morphology and Mechanical Properties in A Murine Model
   Role: Mentor
   Amount: MYR 51,200 (~USD 12,360)
   Duration: 2 years (September 2019 – August 2021) [**Completed**]
3. Fundamental Research Grant Scheme (FRGS)
   Topic: The elucidation of molecular toxicity of imaging-based metallic nanoparticles contrast media
   Role: Member
   Amount: MYR 134,200 (~USD 32,000)
   Duration: 2 years 9 months (September 2019 – May 2022) [**Completed**]
4. Fundamental Research Grant Scheme (FRGS)
   Topic: Fundamental study of CMOS with in-pixel intelligence in three-dimensional optical computed tomography dosimetry system developed in-house
   Role: Member
   Amount: MYR 158,810 (~USD 39,560)
   Duration: 3 years (September 2019 – August 2022) [**Completed**]
5. Prototype Research Grant Scheme (PRGS)
   Topic: A smart dispenser prototype development for dispensing high-dose radiopharmaceutical in nuclear medicine
   Role: Member
   Amount: MYR 76,000 (~USD 18,350)
   Duration: 2 years (August 2019 – July 2021) [**Completed**]
6. Fundamental Research Grant Scheme (FRGS)
   Topic: Watermelon Branched Chain Amino Acids Status as Essential Muscle Nutrients for Health Living
   Role: Member
   Amount: MYR 104,000 (~USD 26,000)
   Duration: 2 years (August 2016 – July 2018) **[Completed]**
7. Fundamental Research Grant Scheme (FRGS)
   Topic: Advanced Muscle Strength Assessment of Functional Electrical Stimulation Using Non-ionising Radiation Approach
   Role: Member
   Amount: MYR 61,000 (~USD 15,000)
   Duration: 2 years (August 2016 – July 2018) **[Completed]**
8. Research Acculturation Grant Scheme (RAGS)
   Topic: Synthesizing and Mapping Conjugated Gold Nanoparticles for Brain Tumor
   Role: Member
   Amount: MYR 52,000 (~USD 13,000)
   Duration: 2 years (December 2015 – November 2017) **[Completed]**
9. Research Acculturation Grant Scheme (RAGS)
   Topic: Haemodynamic modelling of 2-deoxy-2-(18F)fluoro-D-glucose
   Role: Member
   Amount: MYR 80,000 (~USD 22,000)
   Duration: 2 years (December 2014 – November 2016) **[Completed]**

10. Research Acculturation Grant Scheme (RAGS)
    Topic: Experimental and numerical study of thermo fluid properties of porous copper samples as heat sinks
    Role: Member
    Amount: MYR 80,000 (~USD 22,000)
    Duration: 2 years (December 2014 – November 2016) **[Completed]**
11. Principal Investigator Support Initiative (PSI)
    Topic: The Body Excretion of Metallic Contrast Media (Part 2)
    Role: Project Leader
    Amount: MYR 60,000 (~USD 19,000)
    Duration: 2 years (December 2013 – November 2015) **[Completed]**
12. Principal Investigator Support Initiative (PSI)
    Topic: DNA Gycosylase Activities Induced by Low Dose Ionizing Radiation (Part 2)
    Role: Member
    Amount: MYR 60,000 (~USD 19,000)
    Duration: 2 years (December 2013 – November 2015) **[Completed]**
13. Research Acculturation Grant Scheme (RAGS)
    Topic: The Body Excretion of Metallic Contrast Media
    Role: Project Leader
    Amount: MYR 66,000 (~USD 21,000)
    Duration: 2 and half years (December 2012 – June 2015) **[Completed]**
14. Research Acculturation Grant Scheme (RAGS)
    Topic: DNA Gycosylase Activities Induced by Low Dose Ionizing Radiation
    Role: Member
    Amount: MYR 54,000 (~USD 17,000)
    Duration: 2 and half years (December 2012 – June 2015) **[Completed]**

Private/International:

1. Gilead Sciences Incorporation
   Topic: An open-label, clinical feasibility study of the efficacy of Remdesivir for Long Covid (ERASE-LC)
   Role: Co-investigator
   Amount: £1,254,076.62
   Duration: 1 year (September 2024 – September 2025) [**Completed**]
2. Malaysia Productivity Corporation (MPC)
   Topic: Modeling Phenolic and Sugar Profiles of Watermelons as Novel Maturity Indicators
   Role: Project Leader
   Amount: MYR 101,500 (~USD 24,300)
   Duration: 1 year and 9 months (April 2019 – December 2020) [**Completed**]

INNOVATION AWARDS:

1. Awarded with a gold medal at Virtual Materials Technology Challenges 4.0 (v-MTC 4.0) 2020 organised by The Malaysian Solid State Science and Technology Society (MASS Chapter UPM), Department of Physics, Faculty of Science, UPM.
2. Awarded with a silver medal at Invention, Innovation, Design and Expo 2019 (IIDEX 2019) organised by Universiti Teknologi MARA (UiTM).
3. Awarded with a silver medal at Invention, Innovation, Design and Expo 2018 (IIDEX 2018) organised by Universiti Teknologi MARA (UiTM).

4. Awarded with a silver medal at Invention, Innovation, Design and Expo 2016 (IIDEX 2016) organised by Universiti Teknologi MARA (UiTM).
5. Awarded with a bronze medal at Invention, Innovation, Design and Expo 2016 (IIDEX 2016) organised by Universiti Teknologi MARA (UiTM).
6. Awarded with a bronze medal at Invention, Innovation, Design and Expo 2014 (IIDEX 2014) organised by Universiti Teknologi MARA (UiTM).
7. Awarded with a gold medal at Invention, Innovation, Design and Expo 2013 (IIDEX 2013) organised by Universiti Teknologi MARA (UiTM).
8. Awarded with bronze medal/certificate at 2011 Invention, Innovation & Design Competition for the UiTM Puncak Alam and Puncak Perdana Campus (IID D'Puncak 2011).
9. Awarded with bronze medal at 2010 Pameran Penyelidikan, Rekacipta & Inovasi of University Putra Malaysia (PPRI 2010).

ACHIEVEMENT AWARDS:

1. The Dean of Postgraduate Research Commendation for Excellence in Supervision 2026, awarded by the University of Exeter
2. Best Oral Prize for Radiopharmacy Category at the British Nuclear Medicine Society (BNMS) Annual Spring Meeting 2026.
3. Excellence Service Award for year 2020, awarded by University Putra Malaysia (UPM).
4. Excellence Service Award for year 2016, awarded by University Technology MARA (UiTM).
5. Excellence Service Award for year 2013, awarded by University Technology MARA (UiTM)